Qn. Set Code-7(SOT)

Semester: 5th
Programme: B.Tech
Branch: Comp. Engg.

AUTUMN END SEMESTER EXAMINATION-2025

5rd Semester B.Tech

COMPUTATIONAL INTELLIGENCE

CS30011 / CS 3031

(For 2023 & Previous Admitted Batches)

Time: 2 Hours 30 Minutes | Full Marks: 50

Answer any FIVE questions.
Question paper consists of two SECTIONS i.e. A and B.
Section A is compulsory.
Attempt any Four question from Sections B.
The figures in the margin indicate full marks.
Candidates are required to give their answers in their own words as far as practicable and all parts of a question should be answered at one place only.

SECTION-A

1. Answer the following questions. [1 × 10]

(a) Discuss difference between Membership function and Probability.

(b) Define the following terminologies of fuzzy set:

I) Core, and II) Alpha-cut.

(c) Differentiate between supervised learning and unsupervised learning.

(d) The input to a single-input neuron is 5.0, its weight is 1.5 and its bias is -2. What is the net input to the transfer function?

(e) What is elitism in GA?

(f) Differentiate between Artificial Intelligence and Computation Intelligence.

(g) Explain cylindrical extensions of fuzzy sets.

(h) What is McCulloch-Pitts Neuron?

(i) For the following fuzzy sets, obtain Cartesian product and Cartesian co-product:

A = 0.5/x1 + 0.4/x2 + 0.6/x3 + 1.0/x4

B = 0.7/y1 + 0.8/y2 + 0.1/y3

(j) Suppose that fuzzy set A is described by $\mu_A(x)=bell(x;a,b,c)$. Show that the classical fuzzy complement of A is described by $\mu_{\bar{A}}(x)=bell(x;a,-b,c)$.

SECTION-B

2. (a) Let X = {a, b, c, d}, and Y = {1, 2, 3, 4} and fuzzy sets [5]

A = {(a,0.2),(b,0.5),(c,0.8),(d,1.0)},

D = {(a,0.6),(b,0.9),(c,0.3),(d,0.1)}

B = {(1,0.3),(2,0.9),(3,1.0),(4,0.4)},

C = {(1,0.7),(2,0.4),(3,0.5),(4,1.0)}

Determine the implication relation for:

R="If (x is A AND x is D) then y is B else y is C" using Zadeh's min–max Rule.

(b) Discuss the significance of activation function in Neural Networks. Explain the type of activation functions. In which scenario is the ReLU activation function most suitable? [5]

3. (a) Show the difference between Crisp Set Theory and Fuzzy Set Theory. Discuss in terms of different set operation. Illustrate the difference between Hard computing and Soft computing. [5]

(b) Explain any five one-dimensional parameterized MFs. [5]

4. (a) A fuzzy set J on R is defined as [5]

$$\mu_J(x) = \frac{1}{(1 + (x+2)^4)}$$

Find analytically:

1. Core
2. Support
3. Crossover points
4. Bandwidth
5. Expression for α-cut as a function of α

(b) Consider fuzzy sets representing **"Warm"** and **"Hot"** temperature on the universe U={15,20,25,30,35,40,45}. [5]

"Warm"={(20,0.2),(25,0.5),(30,0.7),(35,0.9),(40,0.6),(45,0.3)}

"Hot"= {(25,0.2),(30,0.6),(35,0.8),(40,1.0),(45,0.9)}

Evaluate the followings:

(i) Union and intersection between these two sets.

(ii) Compliment of both the sets.

(ii) {Warm}-{Hot}

5. (a) Given X= {x1, x2} Y = {y1, y2, y3} Z = {z1, z2, z3} [5]

Let, R1 is the fuzzy relation between X and Y.

Let, R2 is the fuzzy relation between Y and Z.

$$R_1 = \begin{bmatrix} 0.5 & 0.4 & 0.6 \\ 0.7 & 0.8 & 0 \end{bmatrix} \quad R_2 = \begin{bmatrix} 0.2 & 0.6 & 0.1 \\ 0.4 & 1 & 0 \\ 1.0 & 0.3 & 0.8 \end{bmatrix}$$

Find out max-product composition between X and Z (R1 oR2). Make suitable assumptions wherever necessary.

(b) Let the fuzzy set A be defined by a generalized bell MF bell(x, a, b, c). Show that width ($A_{0.99}$) / width ($A_{0.01}$) is a function of parameter b only. Mention and justify the assumptions made, if any. [5]

6. (a) A 2–2–1 neural network (2 inputs, 2 hidden neurons, 1 output neuron) uses the **sigmoid activation function**. Weights and biases are as follows (i=input layer, h=hidden layer and o is output layer): [5]

$$W_{ih} = \begin{bmatrix} 0.2 & 0.4 \\ 0.3 & 0.1 \end{bmatrix}, b_h = [0.5, 0.2]$$

$$W_{ho} = [0.6, 0.9] , b_0 = 0.3$$

For input x= [1,0] and target t=0.5, Learning rate η=0.2.

Perform **one complete iteration** of back-propagation, showing:

- Pictorial representation of the neural model.
- Forward pass values,
- Error term for output neuron,
- Error terms for hidden neurons

(b) Minimize $f(x, y) = x^2 + y^2$ by perforing particle swarm optimization (PSO) for two particles. Parameters: w = 0.5, c1 = 1.2, c2 = 1.2. Run 2 epochs and compare the results of both the epochs. [5]

Initial:

- Particle 1: x1(0) = (−2, 1), v1(0) = (0.5, −0.4)
- Particle 2: x2(0) = (1, 2), v2(0) = (−0.3, 0.1)

Randoms: for both particles r1 = 0.7, r2 = 0.6

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